SURP 2026: MidTerm Report

Note: We are initiating the midterm report submissions to gauge the level of development your projects have gone through from the start.

Instructions:

- Make a copy of this Google Document, complete the necessary fields in the provided form, and have it verified by the professor, including a digital signature.
- Download this document as a **PDF file** (mandatory) and submit it via the designated Google form: Midterm Submission Form
- Deadline: **11.59 PM, 1st July, 2026**
- All students pursuing the SURP Project need to submit the Midterm report duly signed by the professor for any future incentives/resume verification/ certificates.
- In case of any queries, kindly reach out to the EnPoWER Team.

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SURP 2026: MidTerm Report

TO BE FILLED BY THE STUDENT

Details of the student

Name	Kartikya
Roll Number	24B4519
Year of Study	**3rd year**
Contact number	[redacted]

Details of the project

Title of project	Quantitative Research & Trading Framework Project
Name of the professor's guide	Prof. Sandip Mondal (Dept. of Electrical Engineering, IIT Bombay)

Progress Update

Project status: On-going	**Percentage completion (if applicable) :**
List the targets achieved in the project so far	

- Implemented a complete PCA-residual statistical-arbitrage strategy (Avellaneda-Lee framework) end-to-end: PCA factor decomposition, residual extraction, OU mean-reversion fit, and s-score signal generation on a market-neutral long/short book.
- Built a full look-ahead-safe backtesting harness (signal -> decision -> execution -> P&L) with triple-barrier exits, inverse-volatility sizing, dollar-neutrality, and a realistic Indian transaction-cost model (STT, slippage, brokerage).
- Ran the strategy on 10 years of real NSE data (2015-2024, 435 stocks): full-cost result -32.9% (Sharpe -0.56) vs zero-cost +32.9% (Sharpe +0.46) -- isolating a real but thin signal edge erased by costs.
- Statistically validated the edge out-of-sample (Fama-MacBeth t-stats -2.3 to -2.8, significant) and confirmed the harness on known-answer tests; found and fixed a critical P&L bug and avoided two overfitting traps.
- Identified the decisive structural finding via a cost-sensitivity sweep: break-even ~18-20 bps sits BELOW the unavoidable ~22 bps STT floor, proving the strategy is untradable in cash equities but viable in a low-cost regime.
- Froze and documented the cash-equity study, then pivoted to a futures version: verified a

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free NSE F&O bhavcopy data pipeline (continuous-contract stitching + corporate-action adjustment) on a 2-name smoke test.

A summary of the work you have done till now in 7-8 lines

This project develops and tests a PCA-residual statistical-arbitrage strategy for Indian markets. The strategy strips common market/sector factors from stock returns using PCA and trades the mean-reversion of the residual (shorting statistically "rich" names and buying "cheap" ones) in a market-neutral, dollar-neutral book, with triple-barrier exits and inverse-volatility sizing.

I implemented the full pipeline and a look-ahead-safe backtester in Python, then ran it on 10 years of real NSE equity data (2015-2024, 435 stocks) under a realistic Indian cost model. The result is a rigorous negative finding: the signal has a real, statistically significant mean-reversion edge (out-of-sample Fama-MacBeth t-stats -2.3 to -2.8; zero-cost Sharpe +0.46), but it is economically thin and completely erased by transaction costs (full-cost Sharpe -0.56).

A cost-sensitivity analysis pinpointed the cause: the strategy breaks even only around 18-20 bps round-trip, which sits below the ~22 bps Securities Transaction Tax floor alone -- so it is structurally untradable in cash equities under any execution assumption, though it turns modestly profitable in a low-cost (futures-like) regime. Along the way I found and fixed a critical P&L-accounting bug and avoided two overfitting traps, and verified all results through independent equity reconstruction and known-answer tests.

Having frozen and documented this result, I am now extending the identical strategy to single-stock futures (where shorting is realistic and costs are ~4x lower), using free NSE F&O bhavcopy data. I have already verified the futures data pipeline -- contract stitching and corporate-action adjustment -- on a two-name prototype.

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Add pictures / drive links of the experimental set-up/simulations/literature review texts to support the above, if applicable
SURP_Kartikya

Plan Ahead:

Estimated time to project completion (if applicable) :	Approx. 1-2 weeks (target: mid of July 2026)
Provide a brief month-wise plan of how you plan to meet the above deadline	
• Early July 2026: Scale the verified futures data pipeline to the full F&O universe (~190	

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single-stock futures) -- apply corporate-action adjustment, volume-unit normalization across the mid-2024 format change, and build a point-in-time F&O membership list to avoid survivorship bias.

- Mid July 2026: Port the PCA-residual strategy onto the continuous futures series with a fully realistic Indian futures model -- expiry-based contract rolling and roll costs, lot-size constraints, futures transaction costs, and margin. Re-validate with known-answer and out-of-sample tests.Run the definitive futures backtest (full-cost vs zero-cost, all metrics), compare against the cash-equity benchmark to test whether the lower-cost futures regime makes the edge tradable, and document the final results and conclusions.

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TO BE FILLED BY THE PROFESSOR

How has the performance of the student been so far?	**Satisfactory**
Any comments/suggestions/feedback	
Signature of the professor with date	Digital Signature Sandip Mondal (10001970) 01-Jul-26 03:20:58 PM

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